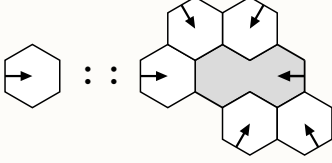


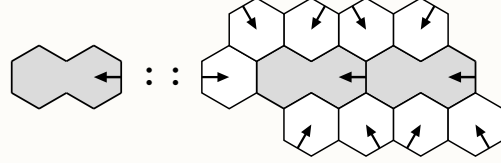
Mountain and Valley

a hat tiling variant

Mountain (H)



Valley (D)



Address Φ

$$\begin{aligned} D_i &\mapsto D_i K_i \\ K_i &\mapsto D_i K_i H_i \\ H_i &\mapsto D_i K_i H_i \\ a_i &\mapsto D_{i-2} a_i H_i \\ b_i &\mapsto D_{i-1} b_i H_i \\ g_i &\mapsto g_i H_i \\ h_i &\mapsto h_i H_i \end{aligned}$$

indices are cyclic modulo 6.

Material

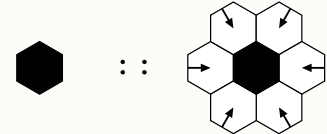
$$\begin{aligned} H_i @ X &\mapsto \begin{cases} D_i @ (\Phi(X) + \langle 0 \rangle), \\ H_j @ (\Phi(X) + \langle D_i K_i, D_i a_{i+2}, D_i b_{i+1}, \\ g_{i-1}, h_{i-2} \rangle); \end{cases} \\ D_i @ X &\mapsto \begin{cases} D_i @ (\Phi(X) + \langle 0, D_i K_i \rangle), \\ H_j @ (\Phi(X) + \langle D_i K_i D_i K_i, D_i a_{i+2}, \\ D_i b_{i+1}, D_i K_i D_i a_{i+2}, D_i K_i D_i b_{i+1}, \\ g_{i-1}, h_{i-2}, D_i K_i g_{i-1}, D_i K_i h_{i-2} \rangle). \end{cases} \end{aligned}$$

j follows the final step direction.

Axioms

name	specification
Single dimer	$D_i @ 0$
Single hexagon	$H_i @ 0$
Fixed-point seed	$D \mid H$
Branch C_3	$\langle H_4 @ e, H_0 @ D_1 K_3, H_2 @ D_2 K_4 \rangle$
Leaf C_3	$\langle H_3 @ e, H_5 @ D_0 K_2 K_3, H_1 @ D_1 K_3 K_4 \rangle$
False center	$C @ 0$

False center



$$C @ 0 \mapsto C @ 0 + H_i @ \langle g_i : i = 0, \dots, 5 \rangle.$$

Two inflations of the $D \mid H$ axiom

